

Pour Over Coffee Brew SOP (1x Cup)

One episode brews one cup of pour over coffee. The dripper sits on the cup for the whole brew, so the coffee runs straight into the cup and is never poured from one vessel to another.

The right gripper does every step. The left arm stays at home for the whole episode and never enters the workspace. Nothing is ever handed between grippers, and only one object is ever held. The right arm is the working arm because the rig is set up right handed, so the dominant hand drives it in leader follower. A left handed collection requires a separately approved mirrored variant that swaps the arm assignments and workspace. Do not mirror this SOP or swap arms during an episode.

The order never changes: **seat the filter, dose one scoop of grounds, bloom, pour one spiral, let the cone drain, stand the dripper on its spot, then bin the spent filter.** Every zone sits in one arc within the right arm's reach, running from the supplies on the inboard side, through the cup in front of the arm, out to the tray and the bin on the outboard side. The scoop and kettle return inboard to their homes after use. Once the wet dripper or spent filter moves outboard, it never travels back over the cup or supplies.

Never touch hot water and never grip the kettle anywhere but the middle of its handle. Never press, stir, or tamp the grounds, and never squeeze the spent filter.

The table is set up in one of three ways. Only the filter stack moves; the cup and dripper, the dripper tray, the bin, the grounds bowl and scoop rest, and the kettle stand are in the same place in all three.

- **Config M:** the filter stack is front-centre, inboard of the cup between the cup and the grounds bowl.
- **Config R1:** the filter stack is front-right, in front of the dripper tray.
- **Config R2:** the filter stack is back-right, behind the dripper tray.

Where a step depends on the setup it says so on an **IF** line — look at the table and follow the line that matches.

What stays constant across all sessions:

- **Start position:** the filter stack starts front-centre (**Config M**), front-right (**Config R1**), or back-right (**Config R2**). One config per episode, chosen before recording and never changed mid-episode.
- **Working arm:** the right gripper takes the filter in every config and the left arm stays at home, so no left zone is used and the filter stack is never placed on the left side of the table. The config changes only the direction of the first reach. Nothing is handed over.
- **Fixed roles:** the cup and dripper stay directly in front of the arm, the tray and the bin stay outboard of the cup, and the grounds bowl, scoop rest, and kettle stand stay inboard of it.

Setup

Complete both checklists before starting an episode.

Cell configuration

- **Environment camera:** 900 mm.
- **cell_type:** bimanual

Hardware checklist

1. Cameras are on and recording.
2. The environment camera shows the whole arc: the kettle stand and grounds bowl on the inboard side, the filter stack in the zone for this episode's config, the cup and dripper in front of the arm, and the dripper tray and bin outboard.
3. The kettle spout and the inside of the filter cone stay visible for the whole pour, and the bed surface stays visible until the cone has drained.
4. The body of the cup stays visible below the dripper, so the rising coffee level can be seen.
5. The whole mouth of the waste bin stays in view.
6. The right arm is at home with its gripper open.
7. The left arm is at home, clear of the arc, and stays there for the whole episode.
8. The tabletop is clear, dry, and free of anything but the zones below.
9. Every zone, including the filter stack front-centre (Config M), front-right (Config R1), or back-right (Config R2), sits within the right arm's comfortable reach, so no step needs the arm extended to its limit or folded in on itself. Nothing obstructs the arm's path along the arc.

Materials checklist

1. The cup stands empty, dry, and upright on its rubber ring directly in front of the right arm, handle turned toward the arm, and is deep enough to take a full cone of water without the coffee reaching its rim.
2. The dripper is already seated on the cup, empty and dry, level on the rim, its collar clear so the gripper can take it from the side. It sits steadily enough to stay put through the pour without being held.
3. The dripper tray is a shallow tray just outboard of the cup, between the cup and the bin, empty and dry, and steady enough to hold the dripper still while a filter is pulled out of it.
4. The waste bin sits open at the outboard end of the arc, mouth clear and low enough to stay in camera view without the arm having to fold in to reach it.
5. The filter stack holds cone filters nested face up, folded rims clear of the holder so the gripper can take the top one from above, in the zone for this episode's config:
 - **Config M:** front-centre, inboard of the cup between the cup and the grounds bowl.

- **Config R1:** front-right, in front of the dripper tray, clear of the path from the tray to the bin.
 - **Config R2:** back-right, behind the dripper tray, clear of the path from the cup to the tray.
6. The grounds bowl stands open inboard of the cup, between the kettle stand and the cup, holding enough for one full scoop with room for the scoop to enter and be drawn up the inside wall. The scoop stands in its rest beside it, handle up, clean and dry.
 7. The kettle is closed and seated on its stand at the inboard end of the arc, handle toward the arm, holding enough water at brewing temperature for one bloom and one spiral with water left over.

Workspace layout

One arc, six zones. Reading from the inboard end (toward the middle of the table) to the outboard end:

1. **Kettle stand** (inboard end)
2. **Grounds bowl and scoop rest**
3. **Filter stack**, front-centre between the grounds bowl and the cup (Config M), front-right in front of the tray (Config R1), or back-right behind the tray (Config R2).
4. **Cup and dripper**, directly in front of the arm on the rubber ring. This is the work point and the arm returns to it every step.
5. **Dripper tray**, just outboard of the cup. Empty until Step 5.
6. **Waste bin** (outboard end)

The grounds bowl and the kettle are inboard of the cup and waste is outboard of it, and in Config R1 and R2 the filter stack sits beside the tray, clear of the arc, so the wet dripper and the spent filter never travel back over the filter stack, the grounds bowl, or the kettle.

Vocabulary

- **Arm assignment:** the right gripper performs every action in this SOP, and takes the filter from the inboard side in Config M or from the outboard side in Config R1 and R2. The left arm never leaves home. Only one object is held at a time, and nothing is handed between grippers.
- **Start zone:** where the filter stack stands at the start of the episode — front-centre (**Config M**), front-right (**Config R1**), or back-right (**Config R2**). One per episode, chosen before recording and never changed mid-episode.
- **Tool home:** the scoop's home is its rest, the kettle's is its stand. Each goes back to its home before anything else is picked up, and neither is ever set down anywhere else.
- **Release point:** the gripper opens only inside the filter cone, over the bin, onto the dripper tray, into the scoop rest, or onto the kettle stand. Never over the grounds bowl, the cup, or

open tabletop.

- **Seated filter:** one filter, open against the cone wall, folded rim above the dripper rim, no corner tented and no fold collapsed inward.
- **Level scoop:** filled inside the bowl and drawn up its inside wall so the excess falls back in, not lifted out heaped.
- **Bloomed bed:** all the grounds are wet, with no water pooling on top.
- **Spiral pour:** one unbroken stream from the middle of the bed, outward in a widening spiral short of the filter wall, and back in to the middle.
- **Spout gap:** a visible gap between the spout and the filter, the bed, and the dripper wall, held for the whole pour.
- **Pour full:** the water has risen near the top of the filter without reaching the folded rim, or the coffee has risen near the cup rim. Whichever comes first ends the pour.
- **Drained cone:** the pool above the grounds has gone and the bed surface is exposed and flat. Read it off the bed inside the cone, not off the spout, which is hidden inside the cup. The cone does not have to stop dripping.
- **Unheld stack:** from setup through the end of draining in Step 4, the dripper and cup are never gripped, steadied, or nudged. The rubber ring holds the cup and the dripper sits on it under its own weight. Step 5 then permits the dripper, but not the cup, to be gripped by its collar and moved to its tray. If either moves out of place before Step 5, end the episode and correct the setup during reset; do not reseal or steady the stack while recording.
- **Stable placement:** the object stands still after release and does not rock, lean, or slide.
- **Spill:** grounds or water on the tabletop. Left where they fall and cleared in the reset, never on camera. Hot water follows the Step 3 spill branch.

Steps

Every step below is performed by the right gripper. Only the filter pick in Step 1 depends on the config: the filter is carried outboard to the dripper in Config M and inboard to it in Config R1 and R2. Every other line is the same in all three configs.

Step 1: Seat the filter

Goal: one filter sits open against the cone wall.

Look where the filter stack is before reaching for the filter.

- **IF the filter stack is front-centre (Config M):** take the top filter by its folded rim, lift it straight up, and carry it outboard to the dripper.
- **IF the filter stack is front-right (Config R1):** take the top filter by its folded rim, lift it straight up, and carry it inboard, clear of the tray, to the dripper.

- **IF the filter stack is back-right (Config R2):** take the top filter by its folded rim, lift it straight up, and carry it forward and inboard, clear of the tray, to the dripper.

Then, in all three:

- Lower it into the cone until it sits against the wall, and release.

Take the filter by its rim only. Never reach inside the cone.

Check: the filter is seated. If it is crooked, tented, or doubled, lift it out and reseal it once. A doubled filter goes in the bin, not back on the stack.

Step 2: Dose one scoop of grounds

Goal: one level scoop sits in the filter and the scoop is back in its rest.

- Take the scoop from its rest.
- Lower it inside the grounds bowl, fill it, and draw it up the inside wall so the excess falls back in.
- Carry it level to the dripper, lower it inside the filter rim, and tip it until it is empty.
- Return the scoop to its rest and release.

One scoop only. No second scoop and no topping up a short one. Never tip the scoop above the filter rim, never touch the grounds with the gripper, and never press, stir, or tamp the bed.

Check: the grounds are in the filter and the scoop is in its rest. If the scoop pulled the filter out of shape, reseal it once.

Step 3: Bloom, then pour one spiral

Goal: the bed is bloomed, one spiral is poured, and the kettle is back on its stand.

The dripper is not held at any point in this step. It sits on the cup under its own weight while the gripper works the kettle.

3.1 Bloom

- Take the kettle by the middle of its handle and lift it straight off the stand.
- Carry it upright, bring the spout over the middle of the bed at the spout gap, and pour until the bed is bloomed.
- Bring the kettle upright, hold it clear of the dripper, and count 10 seconds.

3.2 Pour the spiral

- Bring the spout back over the middle and make one spiral pour, holding the spout gap throughout.
- Stop the moment the cone is pour full.
- Bring the kettle upright over the dripper and hold for 2 seconds so it stops dripping.

One bloom, one spiral. Never break and restart the stream, never pour down the filter wall or onto the dripper, and never add a third pour.

3.3 Return the kettle

- Carry the kettle upright to its stand, lower it until seated, and release. Never carry it away from the dripper still tilted.

Check: the kettle is on its stand, the dripper is still level on the cup, the filter is still seated, and water is draining through the bed rather than standing over the filter rim.

Spill branch: if hot water leaves the dripper or the cup, seat the kettle if the path is clear, clear the arm, and end the episode. Do not move the dripper or the cup and do not wipe hot water while recording. Use the station emergency procedure if the kettle cannot be returned safely.

Step 4: Let the cone drain

Goal: the pool above the bed has gone.

- Hold the arm clear and watch the water sitting above the bed inside the cone.
- Move on once the cone is drained.

Never lift the dripper while water still stands over the bed, and never tap, shake, or press it to speed the draining up. Waiting for the last drop is not required.

Step 5: Stand the dripper on its tray, then bin the filter

Goal: the dripper stands empty on its tray and the filter is in the bin.

Set the dripper down before touching the filter. Never pull a filter out of a dripper held in the air.

- Take the dripper by the collar and lift it straight up off the cup, keeping it level.
- Carry it outboard to the tray, lower it until it stands, and release. Nothing passes back over the cup again.
- With the dripper standing, take the spent filter by its folded rim and lift it straight up out of the cone, keeping it upright so the grounds stay inside.
- Carry it outboard to the bin, lower it inside the bin mouth, and open the gripper.

Never squeeze the filter, and never lift, tip, or shake the dripper to get it out. If the dripper comes up with the filter, set it back down and take the filter again.

Check: the filter is in the bin, the dripper stands empty on its tray, and the cup is uncovered on its ring. If the dripper rocks on the tray, reseal it once.

Step 6: End the episode

- Confirm the cup holds the brew on its ring, the dripper is empty on its tray, the filter is in the bin, and the kettle and scoop are in their homes.

- Return the right arm home, then stop recording.

After the episode: reset the workspace

This reset is not recorded.

1. Let the kettle and the cup reach a safe handling temperature, then empty the cup.
2. Wash and dry the cup and the dripper, stand the cup on its ring in front of the arm with its handle toward the arm, and seat the empty dripper back on it.
3. Wipe the dripper tray dry and leave it empty.
4. Wash and dry the scoop and stand it back in its rest, handle up.
5. Wipe the kettle exterior, refill it, bring it back to brewing temperature, and seat it on its stand with the handle toward the arm.
6. Top up the grounds bowl to hold one full scoop with room for the scoop, and leave it open. Replace grounds that have clumped or dried out.
7. Restock the filter stack, nested face up with rims clear, and stand it in the start zone for the next episode's config — front-centre (Config M), front-right (Config R1), or back-right (Config R2).
8. Empty the bin if the used filters are stacking near its mouth.
9. Wipe the tabletop and run both Setup checklists again.

SOP violations

These are actions that break the SOP and are reviewed side by side in the review tool.

How to record a violation in review

For each violation, record the **start timestamp**, **violation name**, and **SOP rule broken**. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

Violations

Note on the start position: the violations below were written for Config M (filter stack inboard of the cup, between the cup and the grounds bowl). The pickup and arm-role cues will be rewritten later to cover all three start positions; they are left as they are for now. Until then, anything that does not match the episode's config goes under **Config misaligned**.

Violation: Config misaligned

- **Visible cue:** what the operator does does not match the config on the table — the filter stack is not in the start zone for the config; the filter is carried to the dripper from the wrong side; or the wrong IF line is followed.
- **SOP rule broken:** the start position (the filter stack stands front-centre in Config M, front-right in Config R1, or back-right in Config R2, one config per episode; the IF line followed is the one for the config on the table).
- **Coaching note:** look where the filter stack is before the first reach, then follow that config's IF line in Step 1.

Violation: Left arm used

- **Visible cue:** the left arm leaves home at any point, enters the workspace, takes or touches any object, steadies the dripper or the cup, or is used to help with a step.
- **SOP rule broken:** Steps 1 to 5, the right gripper performs every action and the left arm never leaves home.
- **Coaching note:** one arm does the whole task. The left arm stays parked.

Violation: Required sequence not followed

- **Visible cue:** grounds go in before the filter is seated, the kettle is taken before the dose is done, the dripper is lifted before the cone has drained, the filter is pulled before the dripper is standing on its tray, or a step is skipped or repeated.
- **SOP rule broken:** Steps 1 to 6 (seat the filter, dose, bloom and spiral, drain, stand the dripper on its tray and bin the filter, then end).
- **Coaching note:** filter, grounds, water, wait, set it down, bin.

Violation: Filter seated incorrectly

- **Visible cue:** the filter is tented, folded over on itself, crooked in the cone, or two filters go in together, and the episode continues without a reseal.
- **SOP rule broken:** Step 1, seat one filter open against the cone wall, and reseal once if it is crooked, tented, or doubled.
- **Coaching note:** look into the cone before you reach for the scoop.

Violation: Wrong dose

- **Visible cue:** a second scoop is taken, a partial scoop tops up a short one, or the scoop is taken without drawing it up the bowl wall so it goes in heaped.
- **SOP rule broken:** Step 2, one scoop only, levelled on the way out of the bowl, with no second scoop and no top up.
- **Coaching note:** one scoop, levelled on the way out.

Violation: Grounds released from height or outside the filter

- **Visible cue:** the scoop tips above the filter rim and the grounds scatter, or grounds land on the dripper, the cup, or the tabletop.
- **SOP rule broken:** Step 2 (lower the scoop inside the filter rim before tipping it).
- **Coaching note:** get inside the cone before you tip.

Violation: Bed pressed, stirred, or tamped

- **Visible cue:** the scoop, the spout, or the gripper is used to level, push, stir, or press the grounds or the wet bed.
- **SOP rule broken:** Steps 2 and 4 (never press, stir, or tamp the bed, and never touch the grounds with the gripper).
- **Coaching note:** leave the bed how it lands. Only water moves it.

Violation: Bloom skipped or pause not held

- **Visible cue:** the spiral starts with no separate bloom, the bloom floods the bed with water pooling on top, or the episode moves on well before a 10 second count.
- **SOP rule broken:** Step 3.1, pour until the bed is bloomed, then hold the kettle upright and clear and count 10 seconds.
- **Coaching note:** wet it, stop, count ten. Then pour.

Violation: Wrong pour

- **Visible cue:** the stream breaks and restarts during the spiral, the spout is held over one spot instead of spiralling, water runs down the filter wall or onto the dripper, or a third pour is made.
- **SOP rule broken:** Step 3.2, one spiral pour clear of the filter wall, one bloom and one spiral.
- **Coaching note:** start in the middle, spiral out and back, never break the stream.

Violation: Poured past the filter rim or past the cup

- **Visible cue:** water rises to or over the folded rim of the filter, floats the filter off the cone wall, or the coffee reaches the cup rim and the pour continues.
- **SOP rule broken:** Step 3.2, stop the moment the cone is pour full.
- **Coaching note:** watch the filter rim and the cup. Whichever fills first ends the pour.

Violation: Kettle carried, tipped, or set down incorrectly

- **Visible cue:** the kettle is gripped on its body, lid, or spout; tipped before the spout is over the bed; carried away still tilted; the spout touches the filter, the bed, or the dripper; or it is released before it is seated.
- **SOP rule broken:** Steps 3.1 to 3.3, take the kettle by the middle of the handle, carry it upright, hold the spout gap, and seat it before releasing.

- **Coaching note:** handle only, upright to and from, gap over the bed, seated before you open.

Violation: Dripper or cup steadied

- **Visible cue:** the gripper closes on the dripper collar or the cup to hold it still, rests against either during the pour, or the kettle is set down part way through so the dripper can be held.
- **SOP rule broken:** Step 3, the dripper is not held at any point, it sits on the cup under its own weight.
- **Coaching note:** the ring and its own weight hold the stack. If it moves, return the kettle safely if necessary, end the episode, and correct the setup during reset; never steady it during recording.

Violation: Dripper lifted before the cone drained

- **Visible cue:** the dripper is lifted while water is still standing over the bed, or it is tapped, shaken, or pressed to speed the draining up.
- **SOP rule broken:** Step 4, let the cone drain until the pool above the bed has gone.
- **Coaching note:** look into the cone. Move when the bed comes clear, not when the dripping stops.

Violation: Dripper carried back over the cup or the supplies

- **Visible cue:** the wet dripper or the spent filter is carried inboard across the cup, the filter stack, the grounds bowl, or the kettle after being lifted clear.
- **SOP rule broken:** Step 5, carry the dripper and the filter outboard, and nothing passes back over the cup or the supplies.
- **Coaching note:** once it is off the cup, everything travels outboard to the tray and the bin.

Violation: Dripper not stood on its tray

- **Visible cue:** the dripper is put back on the cup, set on open tabletop, left on the filter stack, held in the gripper through the filter pull, or released rocking on the tray.
- **SOP rule broken:** Step 5 (lower the dripper onto the tray until it stands, then release; reseal once if it rocks).
- **Coaching note:** the dripper goes down on its tray before you touch the filter.

Violation: Filter pulled from a held or tipped dripper

- **Visible cue:** the filter is pulled while the dripper is still in the gripper or off its tray, the dripper is lifted or tipped to get the filter out, it is shaken to loosen the bed, or it comes up with the filter and is not set back down.
- **SOP rule broken:** Step 5 (pull the filter with the dripper standing on its tray; if the dripper comes up with it, set it back down and take the filter again).

- **Coaching note:** set it down first, then lift the filter straight up out of it.

Violation: Filter discarded incorrectly

- **Visible cue:** the filter is squeezed, turned out over the tabletop, carried tipped so grounds fall on the way, or dropped from above the bin so grounds scatter.
- **SOP rule broken:** Step 5 (lift the filter out upright by its folded rim, lower it inside the bin mouth, then open the gripper; never squeeze it).
- **Coaching note:** keep it upright, take it all the way into the bin, then open.

Violation: Tool laid down outside its home

- **Visible cue:** the scoop is set on the tabletop, on the bowl rim, or across the dripper; or the kettle is set anywhere other than its stand.
- **SOP rule broken:** Steps 1 to 5, each tool goes back to its home before anything else is picked up.
- **Coaching note:** every tool has one home. It goes there every time.

Violation: Gripper released over the wrong place

- **Visible cue:** the gripper opens over the grounds bowl, over the cup, or over open tabletop, or a tool is released anywhere other than its home.
- **SOP rule broken:** Steps 1 to 5, the gripper opens only inside the filter cone, over the bin, onto the dripper tray, into the scoop rest, or onto the kettle stand.
- **Coaching note:** know where it lands before you open the gripper.

Violation: Required check, retry, or recovery not followed

- **Visible cue:** a failed check is visible and the episode continues without the written correction, a reseal is done more than once, or a recovery not written in the SOP is used.
- **SOP rule broken:** Steps 1 to 5, perform each check and use only its stated retry or recovery, including the Step 3 spill branch.
- **Coaching note:** stop at each check and use only its written recovery.

Violation: Spill collected mid episode

- **Visible cue:** an arm stops to wipe, sweep, or push spilled grounds or water back into the filter, the cup, or the bowl, or wipes hot water while recording.
- **SOP rule broken:** Step 3, spills are left where they fall, and a hot water spill ends the episode without wiping.
- **Coaching note:** leave grounds or non-hot residue where it falls and continue. For a hot-water spill, return the kettle if safe, clear the arm, and end the episode as required by Step 3.

Violation: Item dropped, collided, or knocked over

- **Visible cue:** the kettle, the scoop, a filter, or the dripper falls or is struck; the cup is knocked off its ring; the bin or the dripper tray is tipped or dragged; or the bowl or filter stack is knocked over.
- **SOP rule broken:** Steps 1 to 5, carry every object level and clear of the rim of anything it passes over, and release only after a stable placement.
- **Coaching note:** check the path and the landing spot before you move.

Violation: Workspace or tool state not restored

- **Visible cue:** the episode reaches Step 6 with the scoop out of its rest, the kettle off its stand, the dripper off its tray, or the filter still in the cone.
- **SOP rule broken:** Steps 2, 3.3, and 5 (return the scoop and the kettle to their homes, stand the dripper on its tray, and put the filter in the bin).
- **Coaching note:** run the final confirmation before you home the arm.

Violation: Wrong episode ending

- **Visible cue:** the episode ends with the cup empty or off its ring, the dripper still on the cup, the arm away from home, the gripper still holding something, or no 3 second hold on the finished cup.
- **SOP rule broken:** Step 6 (confirm the cup, the dripper, the filter, the kettle, and the scoop; return the right arm home; then stop recording).
- **Coaching note:** confirm, hold, home, stop.

Failures that are not violations

These are not caused by how the task was run. Log them as system issues, discard the episode, and never use them for coaching.

- **Recording stopped or paused during the episode** (recording system).
- **Camera dropped frames or lost its feed** (capture system).
- **Hardware fault on an arm:** gripper failure, drift, controller caused collision, or motor error.
- **Kettle fault:** it will not pour a controllable stream, runs dry part way through the spiral, or its water is below brewing temperature at the start.
- **Filter fault:** a filter tears in the gripper, splits during the pour, or the stack is empty.
- **Defective grounds:** the bowl is empty, or the grounds are clumped into a mass the scoop cannot break.
- **Bed will not drain** or the cone floods and stalls with nothing visibly done wrong, so the brew never comes through. Check the grind before the next episode.
- **Dripper will not stay put** on the cup during the pour under its own weight, or will not sit level on the cup rim or on its tray, or the tray will not hold it still while a filter is pulled.
- **Cup fault:** the cup leaves its ring under its own weight, or the ring slides on the tabletop.

- **Scoop or rest fault:** a scoop that will not hold a level dose, or a rest slot that will not hold the scoop upright.
- **A zone shifts out of the arc** during the episode, or a zone turns out to sit outside the arm's comfortable reach so a step cannot be completed without extending or folding the arm.

Annotation subtasks (from SOP)

1. Seat a filter in the dripper
2. Take the scoop from its rest
3. Move one scoop of grounds from the bowl to the filter
4. Return the scoop to its rest
5. Take the kettle from its stand
6. Pour the bloom and hold the pause
7. Pour one spiral over the bed
8. Return the kettle to its stand
9. Let the cone drain
10. Move the dripper from the cup to its tray
11. Move the spent filter from the dripper to the waste bin
12. Return the arm home and end the episode